Stochastic Processes 2021/22 Semester 1

Tutorial 7 Extension Questions

Please note: These questions are an extension to Tutorial 7 for individual training only and are not supposed to be discussed during tutorials or office hours. Answers are available for all questions, but worked solutions are only available for some hard questions.

1. Using the Poisson process properties concluded in Tutorial 7 Question 1, consider the following:

   (a) Assume that during peak hours passengers arrive at a train station as a Poisson process with rate $\lambda = 30$ (per minute). Given that there are 1500 arrivals within the time interval $(5pm, 6pm)$, find the (conditional) probability that there were no arrivals between 5.18 and 5.22.

   (b) Following the assumptions made in (a), find the (conditional) probability that there were 10 arrivals between 5.40 and 5.45.

   (c) Following the assumptions made in (a), assume that each passenger is female with probability 3/5 (and male otherwise) independently of everything else. Find the probability that, within the time interval $(5.40, 5.45)$, four females arrive, given that the total number of arrivals is:

      i. at most 5?

      ii. at least 5 ?

2. Consider the newspaper seller case given in Tutorial 7 Question 2.

   (a) For some busy hours, customers arrive according to a Poisson process with rate 3 per minute. At any given time, what is the probability that at least five minutes have elapsed since the last customer has arrived ?

   (b) Assume the rate at which customers arrive is 1 per minute, and each has probability α of being female. In the case $\alpha = 0.4$, find the probability that more than two female customers arrive in a five-minute period, given that more than four male customers arrive in this period.

(c) Assume the rate at which customers arrive is 1 per minute, but Joe changes his strategy and talks to every fifth customer. What is the distribution of the time (in minutes) between Joe's successive conversations? What is the probability that more than five minutes elapse between any one conversation and the next?

(d) How does this probability change if the rate at which customers arrive increases or decreases?

3. Consider the motorcycle injuries case in Tutorial 7 Question 4, if now male motorcycle injuries occur as a Poisson process of rate 8 per week, while female motorcycle injuries occur as an *independent* Poisson process of rate 4 per week. (Clearly no motorcycle ever carries more than one person.) Consider both processes from a given time (0 say).

   (a) Find the probability that 6 male injuries occur before 5 female injuries (*use the Superposition Theorem and consider the first 10 injuries*),

   (b) Following the assumptions made in Question 4 (b), calculate x under the new injury rates.